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Room air purification unit

Abstract

Disclosed herein is a room air purification unit. The room air purification unit can comprise a filtration apparatus, a blower, a purification chamber, and an outer housing. The filtration apparatus can be configured to cause a filtration of air from a setting in which the room air purification unit is located when the air flows through the filtration apparatus. The blower can be configured to cause the air from the setting to flow through the filtration apparatus and into an inlet of the blower and to expel the air from an outlet of the blower. The purification chamber can be configured to receive the air from the blower and to cause a purification the air when the air flows through the purification chamber. The outer housing can enclose the filtration apparatus, the blower, and the purification chamber.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5997619	12/1998	Knuth	55/385.2	B01D 46/0038
2013/0291735	12/2012	Livchak	165/48.1	F24F 1/0047
2014/0366734	12/2013	Moro Franco	96/60	B60H 3/0078
2023/0270908	12/2022	Lehmann	422/121	B01D 53/323

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of and priority to U.S. Provisional Application No. 63/238,528, filed Aug. 30, 2021, and entitled “ROOM AIR PURIFICATION UNIT,” which is incorporated by reference herein in its entirety.

BACKGROUND

(1) In indoor settings like offices, circulated air can include dust, airborne pathogens, and other airborne particles that can have a negative effect on health. Thus, steps can be taken to ensure that the air is clean and safe. Unfortunately, filtration systems installed in indoor settings are often ineffectual because they operate at low efficiencies, if any filtration systems are installed at all. High-efficiency filtration is rarely present in indoor settings and adding high-efficiency filtration to those settings may not be feasible. In addition, filtration systems may only filter air without purifying and clearing that air.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of the embodiments and the advantages thereof, reference is now made to the following description, in conjunction with the accompanying figures briefly described as follows:

(2) FIGS. 1A-C illustrate examples of perspective views of the room air purification unit,

according to various embodiments of the present disclosure.

(3) FIGS. 2A-C illustrate examples of top views of the room air purification unit, according to various embodiments of the present disclosure.

(4) FIGS. 3A-B illustrate examples of bottom views of the room air purification unit, according to various embodiments of the present disclosure.

(5) FIGS. 4A-C illustrate examples of first side views of the room air purification unit, according to various embodiments of the present disclosure.

(6) FIGS. 5A-B illustrate examples of second side views of the room air purification unit, according to various embodiments of the present disclosure.

(7) FIG. 6 illustrates an example of an isolated perspective view of a control module of the room air purification unit, according to various embodiments of the present disclosure.

(8) FIG. 7 illustrates an example of an isolated perspective view of an ultraviolet module of the room air purification unit, according to various embodiments of the present disclosure.

(9) FIG. 8 illustrates an example of a flow diagram of a process for filtering and purifying air using the room air purification unit, according to various embodiments of the present disclosure.

(10) The drawings illustrate only example embodiments and are therefore not to be considered limiting of the scope described herein, as other equally effective embodiments are within the scope and spirit of this disclosure. The elements and features shown in the drawings are not necessarily drawn to scale, emphasis instead being placed upon clearly illustrating the principles of the embodiments. Additionally, certain dimensions may be exaggerated to help visually convey certain principles. In the drawings, similar reference numerals between figures designate like or corresponding, but not necessarily the same, elements.

DETAILED DESCRIPTION

(11) In the following paragraphs, the embodiments are described in further detail by way of example with reference to the attached drawings. In the description, well known components, methods, and/or processing techniques are omitted or briefly described so as not to obscure the embodiments. As used herein, the “present disclosure” refers to any one of the embodiments described herein and any equivalents. Furthermore, reference to various feature(s) of the “present embodiment” is not to suggest that all embodiments must include the referenced feature(s).

(12) Disclosed herein is a room air purification unit that can efficiently filter and purify air in a room or other setting. The room air purification unit can filter air at a higher efficiency than filtration solutions in typical settings. For example, the room air purification unit can use a high efficiency particulate air (HEPA) filter to continually clean the air. As another example, the room air purification unit can use a lower-efficiency filter in addition to HEPA filter to filter larger particles from air before the air enters the HEPA filter, thereby increasing a usable life of the HEPA filter. The room air purification unit can substantially increase the quantity of air changes in a setting relative to typical settings. The room air purification unit can even surpass the standards set by The American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) for number of HEPA-filtered air changes per hour.

(13) In addition, the room air purification can further treat the filtered air using, for example, ultraviolet radiation and bipolar ionization. Conventional systems, though, are often unable to incorporate filtration, ultraviolet radiation, and bipolar ionization in a single unit. This is due in part to the risks posed by ultraviolet radiation. For example, if a conventional filtration unit incorporated ultraviolet radiation, components in the conventional filtration unit could be damaged by the ultraviolet radiation. Examples of the disclosure can separate the ultraviolet radiation from other components of the room air purification unit to prevent these components from becoming damaged. Additionally, people who are near a conventional filtration unit could be harmed by ultraviolet radiation that escaped the unit at the point through which air flows from the unit. In contrast, the room air purification unit can separate the ultraviolet radiation from the point at which air flows out of the room air purification unit to prevent harm to any persons located proximate to

this point. Thus, the room air purification unit can mitigate the risks posed by ultraviolet radiation while still allowing air to flow through the room air purification unit and be filtered and purified. The room air purification unit can likewise be constructed in a way that optimizes the filtration and treatment using ultraviolet radiation and bipolar ionization, which can allow substantially all of the air that flows through the room air purification unit to be exposed to the ultraviolet radiation and bipolar ionization after that air has been filtered.

(14) The room air purification unit can be a standalone unit that does not need to be connected to a building's heating, ventilation, and air conditioning (HVAC) ductwork. Likewise, the room air purification unit can be self-contained in a package that can be easily incorporated into existing indoor settings. The room air purification unit can also be mobile, which can allow the room air purification unit to be located and re-located to rooms or areas where air purification is desired. This allows the flexibility to select settings in which to use the room air purification unit based on each setting's respective air quality requirements, as well as to move the room air purification unit to different settings if those air quality requirements change. The room air purification unit can provide access to internal components on multiple sides for maintenance and service. In some examples, maintenance can be performed on the room air purification unit without using tools.

(15) FIGS. 1A-C illustrate examples of perspective views of the room air purification unit **100**. FIG. 1A shows an outside of the room air purification unit **100**, while FIGS. 1B and 1C show internal components of the room air purification unit **100**. The room air purification unit can include an outer housing **103**, a top cover **106**, a filtration apparatus **109**, a lower chamber **112**, a separator deck **114**, a purification chamber **115**, and an upper chamber **118**.

(16) The outer housing **103** can enclose internal components of the room air purification unit **100**. In some examples, the outer housing **103** can be made from galvanized steel that can resist corrosion and damage. And in some examples, the outer housing **103** can be coated with a powder coat finish that can withstand corrosive cleaning agents.

(17) The outer housing **103** can include an intake grille **121** and access panels **124**. The intake grille **121** can be a grating, screen, register, or other group of openings through which air can enter the room air purification unit **100**. The access panels **124** can be removed to provide access to internal components of the room air purification unit **100** for maintenance. Casters **127** can be attached to a bottom surface of the outer housing **103** to provide greater mobility of the room air purification unit **100**. In some examples, the casters **127** can be lockable to allow the room air purification unit **100** to remain stationary and prevent accidental or unauthorized movement of the room air purification unit **100**.

(18) The top cover **106** can be a cover placed over an upper portion of the room air purification unit **100** on top of the outer housing **103**. In some examples, the top cover **106** can be made from galvanized steel that can resist corrosion and damage. And in some examples, the top cover **106** can be coated with a powder coat finish that can withstand corrosive cleaning agents. In some examples, the top cover **106** can be removable to provide access to internal components of the room air purification unit **100** for maintenance.

(19) The top cover **106** can include a discharge grille **130**. The discharge grille **130** can be a grating, screen, register, or other group of openings through which air can exit the room air purification unit **100**. Air can exit the room air purification unit **100** through the discharge grille **130** at an upward angle relative to an angle at which air enters the room air purification unit **100**. That way, air that has been purified by the room air purification unit **100** does not immediately re-enter the room air purification unit **100** through the intake grille **121**.

(20) In some examples, the top cover **106** can be sloped relative to the outer housing **103** to deter storage of items on top of the room air purification unit **100**. Items stored on top of the room air purification unit **100** could otherwise block the discharge grille **130**. This can prevent the flow of air out of the room air purification unit **100** from being inhibited.

(21) The filtration apparatus **109** can include one or more filters, combination of filters, or other

filtration means by which air that enters the room air purification unit **100** can be filtered. The filtration apparatus **109** can include, for example, a pre-filter **133** and a filter **136**. The pre-filter **133** and the filter **136** can be any devices designed to remove airborne particulate matter from air or other gasses that flow through them. The filtration apparatus **109** can be positioned proximate to the intake grille **121** so that air is filtered before entering the lower chamber **112**. For example, the pre-filter **133** can be positioned between the intake grille **121** and the filter **136**, and the filter **136** can be positioned between the pre-filter **133** and the lower chamber **112**. The pre-filter **133** can be secured by brackets attached to a bottom surface of the outer housing **103** and to the separator deck **114**. The filter **136** can be secured within a frame that is attached to a bottom surface of the outer housing **103** and to the separator deck **114**. The pre-filter **133** can rest against an outer surface of this frame. The pre-filter **133** and the filter **136** can be accessed for removal and replacement by removing one of the access panels **124**.

(22) In some examples, the pre-filter **133** can be a lower-efficiency filter while the filter **136** can be a higher-efficiency filter. The pre-filter **133** can collect larger airborne particles so that only smaller airborne particles pass into the filter **136**. This can reduce damage to the filter **136** caused by larger particles and allow the filter **136** to be changed less frequently. The pre-filter **133** can be, for example, a low- or medium-efficiency filter, an electrostatic filter, a washable filter, a fiberglass filter, a pleated filter, or any other suitable air filter. A pre-filter **133** can be chosen based on a Minimum Efficiency Reporting Value (MERV) rating sufficient to filter particles large enough to damage the filter **136**. The filter **136** can be, for example, a high-efficiency particulate air (HEPA) filter, an electrostatic filter, a washable filter, a fiberglass filter, a pleated filter, or any other suitable air filter. A filter **136** can be chosen based on a MERV rating needed in the setting in which the room air purification unit **100** will be used. The pre-filter **133** and the filter **136** can also be shaped specially to fit within the room air purification unit **100**.

(23) The lower chamber **112** can be a chamber into which air flows after being filtered through the filtration apparatus **109** and before flowing into the purification chamber **115**. The lower chamber **112** can be located in a lower portion of the room air purification unit **100** below the purification chamber **115** and the upper chamber **118**. An upper boundary of the lower chamber **112** can be defined by a separator deck **114**, and a lower boundary of the lower chamber **112** can be defined by a bottom surface of the outer housing **103**. Lateral boundaries of the lower chamber **112** can be defined by the filtration apparatus **109** and lateral surfaces of the outer housing **103**, including the access panels **124**. The lower chamber **112** can include a blower **139**, a bipolar ionization unit **142**, a control module **145**, a power switch **158**, and an air flow controller **151**.

(24) The blower **139** can create pressure that pulls air through the filtration apparatus **109** and into the lower chamber **112**. The blower **139** can be located in the lower chamber **112** proximate to the filtration apparatus **109** and the bipolar ionization unit **142**. A motor can cause an impeller within the blower **139** to spin and pull air into one or more inlets on the sides of the blower **139**. The impeller can then expel the air from an outlet at a top of the blower and into the purification chamber **115**. The blower **139** can be accessed for maintenance by removing one of the access panels **124**.

(25) While in FIGS. **1B**, **1C**, **2C**, **3B**, **4B**, **4C**, and **5B** the blower **139** is a centrifugal blower, the blower **139** can be any system designed to push and pull air. For example, the blower **139** can be a centrifugal blower, a high-speed blower, an axial fan, a backwards inclined fan, a plug fan, or any other suitable type of blower. The blower used can depend on the components used in the filtration apparatus **109**. For example, if the filter **136** used in the filtration apparatus **109** is a HEPA filter, a blower **139** can be used that is designed to intake air at a velocity suitable for moving that air through the filter **136**.

(26) The bipolar ionization unit **142** can treat the air that has been filtered through the filtration apparatus **109**. The bipolar ionization unit **142** can be any device designed to produce ions that can cluster around airborne pathogens and cause other airborne particles to clump together. As an

example, the bipolar ionization unit **142** can do so by creating and emitting positive and negative oxygen ions without also creating ozone. In some examples, a bipolar ionization unit **142** can be used that requires little-to-no servicing or maintenance.

(27) The bipolar ionization unit **142** can be located in the lower chamber **112** and secured to the blower **139** or other surface in the lower chamber **112**. The bipolar ionization unit **142** can be secured using, for example, a mounting bracket. The bipolar ionization unit **142** can be located proximate to the blower **139** so that ions produced by the bipolar ionization unit **142** are immediately taken into the blower **139** at a high velocity. This can minimize collisions between ions and ensure a high concentration of ions in the air, as well as preventing ions from being caught in the filtration apparatus **109**. In some examples, however, the bipolar ionization unit **142** may be omitted, or another component for treating the air may be used in its place.

(28) The control module **145** can include various components used to control operation of the room air purification unit **100**. The control module **145** can be secured to an inner surface of the outer housing **103** within the lower chamber **112**. In some examples, however, the control module **145** may be located outside of the lower chamber **112** and secured to an outer surface of the outer housing **103**.

(29) The power switch **148** can be any electrical device designed to activate and deactivate the room air purification unit **100**. The air flow controller **151** can be any device used to control a speed of airflow in and out of the room air purification unit **100** once the room air purification unit **100** has been activated using the power switch **148**. The air flow controller **151** can control a speed at which the motor of the blower **139** operates. In some examples, the air flow controller **151** can allow for selection of one or more set speeds. In other examples, the air flow controller **151** can allow for selection of a variable speed. The air flow controller **151** can in some examples be controlled by a computing device such that the computing device can control the airflow speed of the room air purification unit **100**. The air flow controller **151** can in some examples be automated to set an airflow speed of the room air purification unit **100** based on conditions like a size of the setting in which the room air purification unit **100** is located.

(30) The power switch **148** and the air flow controller **151** can each be secured within the lower chamber **112** to an inner surface of the outer housing **103**. The power switch **148** and the air flow controller **151** can be secured proximate to an aperture in the outer housing **103** to provide access to the power switch **148** and the air flow controller **151** from outside of the room air purification unit **100**. In some examples, the power switch **148** and the air flow controller can be connected to the control module **145**.

(31) The separator deck **114** can separate the lower chamber **112** from the purification chamber **115** and the upper chamber **118**. The blower **139** can expel air from the lower chamber **112** and into the purification chamber **115** through an aperture in the separator deck **114**. The separator deck **114** can be secured to lateral surfaces of the outer housing **103**.

(32) The purification chamber **115** can be a chamber in which air that has been filtered can be purified. The purification chamber **115** can include a purification chamber housing **154**, baffles **157**, an ultraviolet lamp **160**, and an ultraviolet module **163**. The purification chamber **115** can be located in an upper portion of the room air purification unit **100** above the lower chamber **112**. Boundaries of the purification chamber **115** can be defined by the separator deck **114** the purification chamber housing **154**, and one or more of the baffles **166**.

(33) The purification chamber housing **154** can enclose the ultraviolet lamp **160** and contain air within a space sufficient to allow substantially all of the air to be treated by the ultraviolet lamp **160**. The purification chamber housing **154** can be positioned over the aperture in the separator deck **114** so that air expelled from the blower is contained within the purification chamber **115** and treated by the ultraviolet lamp **160** before the air leaves the purification chamber **115**. This design also prevents items dropped into the room air purification unit **100** through the discharge grille **130** from falling into the blower **139** and causing damage. In some examples, the purification chamber

housing **154** can be made from, for instance, galvanized steel.

(34) The baffles **157** can be any panels or other devices that can obstruct and direct the flow of the air. The baffles **157** can be connected to or located proximate to the purification chamber housing **154** and angled in a way that directs air out of the purification chamber **115** while containing the air within the purification chamber **115** for sufficient time to allow the air to be treated by the ultraviolet lamp **160**. In some examples, the baffles **157** can be made from, for instance, galvanized steel.

(35) In some examples, inner surfaces of the purification chamber housing **154** and the baffles **157** can be reflective. This reflectivity can increase the intensity of ultraviolet radiation within the purification chamber **115**. The increased intensity of the ultraviolet radiation can allow the air to be sufficiently treated by the ultraviolet lamp **160** during the time that the air is within the purification chamber **115**. The purification chamber housing **154** and the baffles **157** can be made from a reflective material, or a reflective material can be applied to the inner surfaces of the purification chamber housing **154** and the baffles **157**.

(36) The purification chamber housing **154**, the baffles **157**, and the separator deck **114** can together prevent ultraviolet radiation from escaping the purification chamber **115**. This can prevent the ultraviolet radiation from harming any persons present in the setting in which the room air purification unit **100** is located. This can also prevent the ultraviolet radiation from harming other components of the room air purification unit **100** like, for example, the pre-filter **133**, the filter **136**, or bipolar ionization unit **142**. The purification chamber housing **154** can prevent ultraviolet radiation from escaping the purification chamber **115** into a setting in which the room air purification unit **100** is located. The baffles **157** can be angled to allow air to flow out of the purification chamber **115** while also preventing ultraviolet radiation from escaping the purification chamber **115** into the setting. The separator deck **114** can prevent ultraviolet radiation from escaping into the lower chamber **112**.

(37) The ultraviolet lamp **160** can treat the air that has been filtered through the filtration apparatus **109** and expelled by the blower **139** into the purification chamber **115**. The ultraviolet lamp **160** can be any device designed to emit ultraviolet radiation capable of inactivating airborne pathogens. For example, the ultraviolet lamp **160** can be a UV-C lamp. The ultraviolet lamp **160** can be secured to an inner surface of the purification chamber housing **154** or to the separator deck **114**. The ultraviolet lamp **160** can be accessed for removal and replacement by removing either of the access panels **124**. In some examples, however, the ultraviolet lamp **160** may be omitted, or another purification component may be used in its place.

(38) The ultraviolet module **163** can facilitate operation of the ultraviolet lamp **160**. For example, the ultraviolet module **163** can include one or more components that allow the ultraviolet lamp **160** to function. The ultraviolet module **163** can be secured to an outer surface of the purification chamber housing **154** proximate to the ultraviolet lamp **160**.

(39) The upper chamber **118** can be a chamber into which air flows from the purification chamber **115** after being purified. The upper chamber **118** can be located in an upper portion of the room air purification unit **100** above the lower chamber **112**. An upper boundary of the upper chamber **118** can be defined by the top cover **106**, and a lower boundary of the upper chamber **118** can be defined by the separator deck **114** and the purification chamber housing **154**. Lateral boundaries of the upper chamber **118** can be defined by lateral surface of the outer housing **103**, including the access panels **124**. Air can pass through the upper chamber **118** and be expelled from the upper chamber **118** through the discharge grille **130**.

(40) As discussed above, the access panels **124** can be removed to provide access to internal components of the room air purification unit **100** for maintenance. For example, either of the access panels **124** can be removed to provide access to the pre-filter **133** and filter **136** so that the pre-filter **133** or the filter **136** can be removed and replaced. As another example, either of the access panels **124** can be removed to provide access to components of the lower chamber **112** like the blower

139, bipolar ionization unit **142**, control module **145**, power switch **148**, or air flow controller **151**. The blower **139**, bipolar ionization unit **142**, control module **145**, power switch **148**, or air flow controller **151** can then be individually removed for maintenance or replaced altogether. As an additional example, the access panels **124** can be removed to provide access to components of the ultraviolet chamber like the ultraviolet lamp **160** and the ultraviolet module **163** for maintenance or replacement.

(41) FIGS. 2A-2C illustrate examples of top views of the room air purification unit **100**. FIG. 2A illustrates an outside of the room air purification unit **100** from a top view. FIG. 2A shows the top cover **106** and the discharge grille **130**. FIGS. 2B and 2C illustrate internal components of the room air purification unit **100** from top views. FIG. 2B shows the internal components with the purification chamber housing **154** present, and FIG. 2C shows the internal components with at least a portion of the purification chamber housing **154** removed. FIG. 2B shows the separator deck **114**, the purification chamber **115**, the upper chamber **118**, the purification chamber housing **154**, the ultraviolet module **163**, and a baffle **157**. FIG. 2C shows the lower chamber **112**, the separator deck **114**, the purification chamber **115**, the upper chamber **118**, the blower **139**, baffles **157**, the ultraviolet lamp **160**, and the ultraviolet module **163**.

(42) FIGS. 3A-B illustrate examples of bottom views of the room air purification unit **100**. FIG. 3A illustrates an outside of the room air purification unit **100** from a bottom view. FIG. 3A shows the outer housing **103** and casters **127**. FIG. 3B illustrates internal components of the room air purification unit **100** from a bottom view. FIG. 3B shows the filtration apparatus **109**, the lower chamber **112**, the pre-filter **133**, the filter **136**, the blower **139**, the control module **145**, the power switch **148**, and the air flow controller **151**.

(43) FIGS. 4A-C illustrate examples of first side views of the room air purification unit **100**. FIG. 4A illustrates an outside of the room air purification unit **100** from a first side view, FIG. 4A shows the outer housing **103**, the top cover **106**, an access panel **124**, and casters **127**. FIGS. 4B and 4C illustrate internal components of the room air purification unit **100** from first side views. FIG. 4B shows the internal component with the purification chamber housing **154** present, and FIG. 4C shows the internal components with at least a portion of the purification chamber housing **154** removed. FIG. 4B shows the filtration apparatus **109**, the lower chamber **112**, the separator deck **114**, the purification chamber **115**, the upper chamber **118**, the pre-filter **133**, the filter **136**, the blower **139**, the control module **145**, the power switch **148**, the air flow controller **151**, and the purification chamber housing **154**. FIG. 4C shows the filtration apparatus **109**, the lower chamber **112**, the separator deck **114**, the purification chamber **115**, the upper chamber **118**, the pre-filter **133**, the filter **136**, the blower **139**, the control module **145**, the power switch **148**, the air flow controller **151**, baffles **157**, the ultraviolet lamp **160**, and the ultraviolet module **163**.

(44) FIGS. 5A-B illustrate examples of second side views of the room air purification unit **100**. FIG. 5A illustrates an outside of the room air purification unit **100** from a second side view. FIG. 5A shows the outer housing **103**, an access panel **124**, casters **127**, and the top cover **106**. FIG. 5B illustrates internal components of the room air purification unit **100** from a second side view. FIG. 5B shows the filtration apparatus **109**, the lower chamber **112**, the purification chamber **115**, the separator deck **114**, the upper chamber **118**, the pre-filter **133**, the filter **136**, the blower **139**, the bipolar ionization unit **142**, the control module **145**, the power switch **148**, the air flow controller **151**, the ultraviolet lamp **160**, and the ultraviolet module **163**.

(45) FIG. 6 illustrates an example of an isolated perspective view of a control module **145** of the room air purification unit **100**. In the example of FIG. 6, the power switch **158** and the air flow controller **151** are connected to the control module **145**. In some examples, however, the power switch **158** and the air flow controller may be separate from the control module **145**.

(46) In the example of FIG. 6, the control module **145** includes a control unit **603**, a transformer **606**, a terminal block **609**, and a fan relay **612**, but the control module **145** can include different components in other examples. The control unit **603**, transformer **606**, terminal block **609**, and fan

relay **612** can be secured to an inside surface of the control module **145**. The control unit **603** can be any microcontroller, circuit board, or other electronic device configured control operation of the various functions of the room air purification unit **100**. The transformer **606** can be any electrical device designed to supply voltage used to operate the motor of the blower **139**. The terminal block **609** can be any device designed to secure wires or any other connections used to operate the motor of the blower **139**. The fan relay **612** can be any electrical device designed to relay power used to operate the motor of the blower **139**.

(47) FIG. 7 illustrates an example of an isolated perspective view of an ultraviolet module **163** of the room air purification unit **100**. The ultraviolet module **163** can include an ultraviolet ballast **703**. The ultraviolet ballast **703** can be any electronic, magnetic, or electro-mechanical device designed to provide voltage sufficient to initiate the emission of ultraviolet radiation from the ultraviolet lamp **160** and to sustain the emission of the ultraviolet radiation during operation of the ultraviolet lamp **160**.

(48) FIG. 8 illustrates an example of a flow diagram of a process for purifying air using the room air purification unit. For purposes of illustration, a perspective view of the room air purification unit **100** is shown, with an access panel **124** and casters **127** not shown and the purification chamber housing **154** and ultraviolet module **163** shown as transparent.

(49) At step **803**, air can enter the room air purification unit **100** through the intake grille **121**. The room air purification unit **100** can be activated using the power switch **148** and the speed of the air intake can be controlled using the air flow controller **151**. When the room air purification unit **100** is activated, air can be pulled into the room air purification unit **100** by the blower **139**. The air can enter from a room or other setting in which the room air purification unit **100** is located.

(50) At step **806**, the air can pass through the filtration apparatus **109**. The air can first pass through the pre-filter **133**, which can filter out larger airborne particles. The air can then pass through the filter **136**, which can filter out smaller airborne particles. The air can be pulled through the pre-filter **133** and the filter **136** by the blower **139**.

(51) At step **809**, the air can enter the lower chamber **112** after passing through the filtration apparatus **109**. The air that enters the lower chamber **112** can be pulled into the blower **139** through an inlet on a side of the blower **139**. The bipolar ionization unit **142** can produce ions that can be pulled into the blower **139** with the air and thereby mix with the air. These ions can cluster around airborne pathogens and cause other airborne particles to clump together. And because the air has already been filtered and because the ions are pulled quickly into the blower **139**, these ions do not become caught in the filtration apparatus **109**, ensuring an optimal concentration of ions in the air.

(52) At step **812**, the air can enter the purification chamber **115** through an aperture in the separator deck **114** after being expelled from the blower **139**. While in the purification chamber **115**, the air can be exposed to ultraviolet radiation produced by the ultraviolet lamp **160**, which is powered using the ultraviolet module **163**. The ultraviolet radiation can inactivate airborne pathogens in the air. The intensity of the ultraviolet radiation can also be increased by reflectivity of inner surfaces of the purification chamber housing **154** and the baffles **157**. Because the purification chamber **115** is enclosed by the separator deck **114**, the purification chamber housing **154**, and the baffles **157**, this ultraviolet radiation does not escape the purification chamber, preventing harm to other components of the room air purification unit and to any persons present outside of the room air purification unit. The air can then flow around the baffles **157** and out of the purification chamber **115**.

(53) At step **815**, the air can enter the upper chamber **118**. By this step, the air has been treated using ions produced by the bipolar ionization unit **142** and ultraviolet radiation produced by the ultraviolet lamp. Thus, the presence of airborne pathogens and other airborne particles in the air can be significantly reduced or eliminated compared to step **803** when the air entered the room air purification unit **100**. The air can then flow out of the upper chamber **118** through the discharge

grille **130**, out of the room air purification unit **100**, and into the room or other setting in which the room air purification unit **100** is located.

Claims

1. A room air purification unit comprising: a filtration apparatus within a lower chamber, the filtration apparatus configured to cause a filtration of air from a setting in which the room air purification unit is located when the air flows through the filtration apparatus; a blower within the lower chamber, the blower configured to cause the air from the setting to flow through the filtration apparatus and into an inlet of the blower and to expel the air from an outlet of the blower, wherein the filtration apparatus is located between the blower and an intake grille; a bipolar ionization unit within the lower chamber, the bipolar ionization unit configured to produce ions that mix with the air; a purification chamber within a purification chamber housing, the purification chamber located above the blower, comprising at least one ultraviolet lamp, and being configured to receive the air from the blower to cause a purification the air when the air flows through the purification chamber, wherein the purification chamber further comprises a plurality of baffles oriented to contain ultraviolet radiation within the purification chamber, wherein the plurality of baffles comprise an exit baffle angled to allow the air to exit the purification chamber into an upper chamber while preventing ultraviolet radiation from escaping the purification chamber into the lower chamber; a separator deck that separates the purification chamber housing from the lower chamber, wherein the separator deck together with the purification chamber housing prevents the ultraviolet radiation from entering the lower chamber; an upper chamber located above the purification chamber receiving air from the purification chamber, wherein the purification chamber housing and the plurality of baffles prevent the ultraviolet radiation from entering the upper chamber and wherein the air from the purification chamber exits through a discharge grille adjacent to the upper chamber; and an outer housing comprising the discharge grille and enclosing the filtration apparatus, the upper chamber, the blower, and the purification chamber, wherein the discharge grille is angled away from the intake grille and sloped relative to a bottom side of the outer housing.

2. The room air purification unit of claim 1, further comprising a control module configured to control an operation of the blower.

3. The room air purification unit of claim 1, wherein the filtration apparatus comprises a first filter and a second filter.

4. The room air purification unit of claim 3, wherein the first filter is positioned adjacent to an entry of the air to the room air purification unit, and the second filter is positioned between the first filter and the purification chamber.

5. The room air purification unit of claim 3, wherein the second filter has a higher efficiency than the first filter.

6. The room air purification unit of claim 1, wherein the purification chamber housing and the plurality of baffles comprise a reflective inner surface, the reflective inner surface causing an increase in an intensity of the ultraviolet radiation within the purification chamber.

7. An apparatus, comprising: a filtration apparatus within a lower chamber, the filtration apparatus configured to cause a filtration of air from an external setting when the air passes through the filtration apparatus; a purification chamber within a purification chamber housing, the purification chamber located above the lower chamber and being configured to receive the air from the lower chamber to cause a purification the air when the air flows through the purification chamber, wherein the purification chamber further comprises a plurality of baffles oriented to contain ultraviolet radiation within the purification chamber, wherein the plurality of baffles comprise an exit baffle angled to allow the air to exit the purification chamber into an upper chamber while preventing ultraviolet radiation from escaping the purification chamber into the lower chamber; an

ultraviolet lamp within the purification chamber, the ultraviolet lamp configured to generate the ultraviolet radiation and cause a treatment of the air when the air passes within a proximity of the ultraviolet lamp; a blower within the lower chamber configured to cause the air to pass through the filtration apparatus and into an inlet of the blower, the blower causing the air to be expelled from an outlet of the blower, through the purification chamber and into upper chamber positioned above the purification chamber; a separator deck that separates the purification chamber housing from the lower chamber, wherein the separator deck together with the purification chamber housing prevents the ultraviolet radiation from entering the lower chamber; an upper chamber located above the purification chamber receiving air from the purification chamber, wherein the purification chamber housing and the plurality of baffles prevent the ultraviolet radiation from entering the upper chamber and wherein the air from the purification chamber exits through a discharge grille adjacent to the upper chamber; and an outer housing comprising the discharge grille and enclosing the filtration apparatus, the upper chamber, the blower, and the purification chamber, wherein the discharge grille is angled away from an intake grille oriented toward an external setting and sloped relative to a bottom side of the outer housing.

8. The apparatus of claim 7, further comprising a bipolar ionization unit positioned within the purification chamber proximate to the blower, the bipolar ionization unit configured to cause a further treatment of the air.

9. The apparatus of claim 7, wherein the filtration apparatus is positioned between the external setting and the blower in a path of the air from the external setting to the inlet of the blower.

10. The apparatus of claim 7, wherein the ultraviolet lamp is positioned within the purification chamber between the blower and the external setting in a path of the air from the outlet of the blower to the external setting.

11. The apparatus of claim 7, further comprising a control module configured to control an operation of the blower.

12. The apparatus of claim 7, wherein the blower comprises a centrifugal blower, a high-speed blower, an axial fan, a backwards inclined fan, or a plug fan.

13. The apparatus of claim 7, wherein the filtration apparatus comprises a pre-filter and a filter, the filter comprising a high-efficiency particulate air (HEPA) filter, and the pre-filter comprising a filter having a lower efficiency compared to the HEPA filter.
